earthUI

User Guide

General Use · Appraisal Features · Market Area Analysis · Complete Reference

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Version	Audience	Last Updated
0.10.0	All Users	July 2026

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CHAPTER 1

Introduction

What Is earthUI?

earthUI is a graphical user interface for the R `earth` package, which implements Multivariate Adaptive Regression Splines (MARS). It runs as a local Shiny application — there is no login, no server, and no accounts. You launch it from R, import a dataset (CSV or Excel), configure your model, and fit it interactively.

The application provides a complete workflow: data import, variable configuration, model fitting, diagnostic plots, variable importance, model equations, and downloadable reports in HTML, Word, or PDF format.

Three Purpose Modes

Every project has a **Purpose**, chosen when the project is created in the Project section of the sidebar. There are three modes:

- **General** — Earth regression for any type of population or dataset. This is the default mode. It provides the full MARS modeling workflow without any domain-specific additions.
- **For Appraisal** — Earth regression tailored for real estate appraisal. Adds features specific to single-property valuation, including subject property handling, special column designations, Residual Constraint Approach (RCA), and Sales Comparison Grid generation.
- **Market Area Analysis** — Earth regression tailored for market area studies. Adds features for analyzing groups of properties in a defined market, with an optional "Skip first row" checkbox for subject property exclusion.

In all three modes, the core modeling engine is identical — you are always fitting an Earth (MARS) model. The purpose setting controls which additional tools and interface elements are available.

The active project determines the purpose. To work in a different mode, open or create a project with that purpose. Switching projects resets earthUI to a clean default state: imported data, model results, tabs, variable configuration, and earth parameters are all cleared. Previously saved settings for the newly opened project (and its purpose) are then restored automatically.

Real Estate–Specific Features

When either **For Appraisal** or **Market Area Analysis** is selected, earthUI activates several features designed for real estate analysis:

- **Special column designations** — Each predictor can be tagged with a special role such as `contract_date`, `dom`, `concessions`, `latitude`, `longitude`, `living_area`, `lot_size`, `actual_age`, `effective_age`, `area`, `site_dimensions`, or `display_only`. These designations control how the column is handled during fitting, output, and Sales Grid generation. See Chapter 6 for the complete reference.
- **Rounding of latitude and longitude** — Columns designated as latitude or longitude are automatically rounded to 3 decimal places to prevent overfitting.
- **Sale Age column** — When a column is designated as `contract_date` and an Effective Date is provided, earthUI computes a `sale_age` column (days between sale date and effective date) and substitutes it as a predictor.
- **RCA computations (appraisal only)** — In appraisal mode, after fitting the model earthUI can compute Residual Constraint Approach (RCA) output, which produces per-comparable adjustments, net/gross adjustment summaries, and an adjusted sale price for the subject property.
- **Sales Comparison Grid (appraisal only)** — Generates a multi-sheet Excel workbook with formulas for comparable adjustments, residual feature breakdowns, and adjusted sale prices. See Chapter 11.

> ***Tip:*** *The General purpose mode works for any dataset — financial, scientific, engineering, or real estate. The appraisal and market modes simply add convenience features for real estate professionals.*

Getting Started

To use earthUI:

1. **Install the package** in R: `install.packages("earthUI")` or install from source.
2. **Launch the application**: run `earthUI::launch()` in R, or from the command line run `Rscript -e 'earthUI::launch()'`. The app opens in your web browser on port 7878. You can also access the app directly by navigating to `http://localhost:7878` in your browser.
3. **Choose or create a project** in Section 1 (the Project picker) — the purpose (General, For Appraisal, or Market Area Analysis) is set when the project is created — then **import your data** from the project's input folder in Section 2. earthUI accepts CSV and Excel files.
4. **Configure variables** — choose your target(s) and predictors, set data types, and assign any special column roles.

5. **Set Earth parameters** — degree, nprune, subset filters, and other `earth()` arguments.
6. **Fit the model** — click "Fit Earth Model" and review the results in the main panel.
7. **Export** — download predictions as Excel, generate reports, or (in appraisal mode) compute RCA adjustments and Sales Comparison Grids.

Settings are saved to the project database when you click the **Save current as default** buttons (in Sections 3 and 4) and are restored automatically when you reopen the project.

CHAPTER 2

MLS Input Data Requirements

For real estate appraisal and market analysis workflows, your input data typically comes from a Multiple Listing Service (MLS) export. This chapter describes the expected file structure and the columns that earthUI can use.

File Format & Structure

earthUI accepts **CSV** and **Excel** (`.xlsx`, `.xls`) files. On import, column names are automatically converted to `snake_case` — for example, "Living SqFt" becomes `living_sqft`, "Contract Date" becomes `contract_date`, and "Sale Price" becomes `sale_price`. This normalization ensures consistent column references throughout the workflow. The CSV separator and decimal mark used during import are determined by the locale settings (see Chapter 3, "Locale & Regional Settings").

Your data file should be a flat table with one row per property and one column per attribute. The first row of the file must contain column headers.

Required Columns for Appraisal Mode

While earthUI works with any set of columns, the full appraisal workflow (RCA adjustments + Sales Comparison Grid) benefits from having the following columns in your MLS export:

Column	Special Type	Purpose
Sale Price	(target)	Response variable for the model
Contract Date	`contract_date`	Used to compute `sale_age` (days from effective date)
Listing Date	`listing_date`	Used with contract date to compute DOM if no DOM column exists
Days on Market	`dom`	Days on market; displayed in Sales Grid
Concessions	`concessions`	Sale concessions; Net SP = Sale Price — Concessions in Sales Grid
Living Area (SF)	`living_area`	Enables per-SF residuals (`residual_sf`, `cqa_sf`)
Lot Size	`lot_size`	Grouped in "Site Size | Dimensions" row in Sales Grid
Site Dimensions	`site_dimensions`	Grouped with lot size in Sales Grid
Latitude	`latitude`	Rounded to 3 dp; used for proximity calculation and Location group
Longitude	`longitude`	Rounded to 3 dp; used for proximity calculation and Location group
Area ID	`area`	Grouped in "Loc: Long | Lat | Area" row in Sales Grid
Actual Age	`actual_age`	Grouped in "Actual Age | Effective Age" row in Sales Grid
Effective Age	`effective_age`	Grouped with actual age in Sales Grid
Parcel Number	(none)	Displayed in APN row of Sales Grid
Listing ID	(none)	Displayed in APN row of Sales Grid
Address	`display_only`	Displayed in Address row of Sales Grid; excluded from model

Spreadsheet column names can be in a foreign language — the "special" names are in English so that the R program can give them special treatment. Otherwise, the given column names show up in the regression models, graphs, and (if doing appraisals) the Intermediate Sales Grid.

Not all columns are required. earthUI adapts — if a column is missing, the corresponding feature is simply omitted. For example, if no `concessions` column is designated, the Net SP row in the Sales Grid shows Sale Price without a concessions deduction. However, for real estate pricing models certain columns are highly recommended to achieve acceptable fit:

1. **Sale Age** — the number of days between the contract sale date and the effective date of the appraisal or analysis. If multi-year sales history is being used, especially for periods over 5 years, `sale_age` often plays a central role in estimating the sale price. In fact it is often so important that without it, earthUI fails to provide any model at all.
2. **Living Area** — also goes by names such as "Living Sqft," "GLA" (gross living area) and so on. This is another leading determinant of sale price.
3. **Total Bath Count** — the total number of full, quarter, half, and 3/4 bathrooms. For example, two full baths and one half-bath would be a value of 2.5.
4. **Garage Bays** or **Garage Area** — the number of garage spaces or the garage square footage.
5. **Lot Size** — the land area of the property, typically in square feet or acres.
6. **Longitude, Latitude,** and if available **Area ID**. Adjustments for these will be combined under a single Location adjustment in the Sales Grid.

Special Column Naming Conventions

earthUI identifies columns by their **special type designation**, not by their column name. You can name your columns anything you like in the MLS export — what matters is that you assign the correct special type in the Variable Configuration table (Chapter 6).

For example, your MLS might export living area as "GLA", "Living SqFt", "liv_area", or "gross_living_area". After import (where it becomes snake_case), you simply designate it as `living_area` in the Special dropdown. earthUI will then use it for per-SF residual calculations and Sales Grid grouping regardless of its original name.

Data Quality & Completeness

- **Missing values (NA)**: Rows with NA values in any predictor or target column are automatically removed before fitting. The `na.action` is always set to `na.fail` internally, with NA removal handled before the call to `earth()`.
- **Date columns**: Dates should be in a format R can parse (e.g., `2025-06-15`, `06/15/2025`, `June 15, 2025`). earthUI auto-detects date columns when at least 80% of values parse successfully.
- **Numeric columns**: Sale price, living area, lot size, concessions, and similar fields must be numeric. If your MLS exports prices with currency symbols or thousands separators (e.g., "$350,000" or "350.000,00"), you may need to clean these before import.
- **Factor columns**: Categorical variables like area ID, style, or condition should contain a manageable number of unique values. earthUI auto-detects factors but you can override the detection.

> ***Tip:*** *Review the T/F/NA column in the predictor table after import (the third number is the NA count). Columns with more than 30% missing values are flagged in red. Consider excluding high-NA columns or cleaning the data before import.*

Subject Row Placement

In **Appraisal** mode, **row 1 must be the subject property**. All remaining rows are comparable sales. The subject's sale price can be left blank (NA) or set to any value — earthUI treats it as NA during fitting regardless.

In **Market Area Analysis** mode, placing the subject in row 1 is optional. If present, check "Skip first row (subject property)" to exclude it from fitting.

In **General** mode, there is no special row handling — all rows are treated equally.

CHAPTER 3

General Purpose Mode

Overview

General Purpose mode is the default when you launch earthUI. It provides the complete MARS modeling workflow for any dataset — not just real estate. You can use earthUI for scientific data, financial analysis, engineering studies, or any regression problem where you want to explore non-linear relationships and interactions between variables.

In General mode, the interface omits the real estate–specific features (special columns, sale age, coordinate rounding, RCA). The sidebar is streamlined to focus on variable selection, parameter configuration, model fitting, and export.

The Sidebar Workflow

The sidebar is organized into numbered, collapsible sections that guide you from data import through export:

1. Project — Choose an existing project or create a new one (purpose, country, and admin-level location). The active project determines where input files are read from and where outputs and logs are written — there is no separate output-folder field to set.

2. Import Data — Select a data file from the active project's input folder (CSV or Excel). For Excel files with multiple sheets, a sheet selector appears. Column names are automatically converted to snake_case.

3. Variable Configuration — Target variable selector (supports multiple targets), effective date, and a predictor table with checkboxes for Include, Factor, and Linear. See Chapter 6 for full details.

4. Earth Call Parameters — All arguments to the `earth()` function: degree, penalty, nk, pruning method, cross-validation, and more. See Chapter 7 for the complete parameter reference.

Execute Prolog Processing — An unnumbered section between 4 and 5 for the optional Prolog/list-column processing step (off by default; enabled via the Settings gear). See "Advanced: Prolog Remarks Processing" in Chapter 6.

5. Fit Earth Model — A **Random seed** field (pre-filled with a random value, with a recent-seeds recall dropdown) followed by the blue **Fit Earth Model** button, which runs the model asynchronously. See Chapter 8 for fitting details.

6. Download Output — Exports predictions, residuals, CQA scores, and per-g-function contributions as an Excel file. The section heading reads "6. Download Estimated Target Variable(s) & Residuals" (general) or "6. Download Estimated Sale Prices & Residuals" (appraisal/market). See Chapter 9.

In **appraisal mode**, further numbered sections appear: **7. Calculate RCA Adjustments & Download** and **8. Generate Sales Grid & Download**. Report generation is in **9. Generate Quarto Report** and **10. Convert Quarto Report**, saved to the project's output folder. See the later chapters.

Main Panel Tabs

After fitting, the main panel provides ten tabs:

Tab	Contents
Data	Preview of imported data. In appraisal mode, split into Subject Property and Comparable Sales tables.
Equation	The fitted model equation displayed in LaTeX/MathJax, showing each basis function and its coefficient.
Summary	Key metrics (R^2, GRSq, GCV, RSS, CV R^2) and a coefficients table.
Variable Importance	Bar chart and ranked table of predictor importance scores.
Contribution	g-function contribution plots. Select a g-function from the dropdown; univariate predictors show line plots, bivariate interactions (degree ≥ 2) show 3D perspective and contour plots.
Correlation	Heatmap of predictor correlations.
Diagnostics	Three plots: Residuals vs Fitted, Normal Q-Q, and Actual vs Predicted.
RCA Adjustments	RCA adjustment output, populated once step 7 (Calculate RCA Adjustments & Download, appraisal mode) has been run; until then the tab shows a notice to run step 7 first.
ANOVA	ANOVA decomposition table showing each basis function's contribution to RSS.
Earth Output	Raw text output from 'earth::summary()', including the pruning pass details.

Settings Persistence

earthUI saves your configuration **only when you click a Save button** — there is no automatic save-as-you-type. Two buttons do the saving:

- **Save current as default** in Section 3 (Variable Configuration) saves the target, the effective date, and the predictor configuration (Include / Factor / Linear / Type / Special).

- **Save current as default** in Section 4 (Earth Call Parameters) saves the earth call parameters and the Allowed Interactions matrix.

Settings are stored in the project database (`projects.sqlite`), scoped per **project and purpose** — so all data files within a project share one configuration, and the same project can have different configurations for General, Appraisal, and Market modes. Saved settings are restored automatically when you reopen the project. **Fitting does not save settings**, so you can fit experimentally without overwriting your saved default.

Dark Mode

Click the moon/sun icon in the upper-right corner to toggle between light and dark themes. The theme preference is saved in local storage and persists across sessions.

Locale & Regional Settings

earthUI supports international number, date, and CSV formatting conventions through a country-based locale system. The **Country** dropdown in Section 1 of the sidebar (below the file upload) selects a preset for 31 supported countries. Each preset configures:

- **CSV separator** — comma (`,`) for US/UK/Japan or semicolon (`;`) for most of Europe, where the comma is used as a decimal mark.
- **Decimal mark** — period (`.`) or comma (`,`).
- **Thousands separator** — comma (US/UK/Japan), period (Germany/Italy/Spain), space (Finland/France/Poland/Baltics/Ukraine/Russia), or apostrophe (Switzerland).
- **Date format** — MM/DD/YYYY (US), DD/MM/YYYY (most of Europe), or YYYY-MM-DD (Sweden/Lithuania/Japan/Canada).
- **Paper size** — Letter (US/Canada/Mexico) or A4 (everywhere else).

Supported Countries

Country	CSV Sep	Decimal	Thousands	Date	Paper
US, Canada (csv)	,	.	,	MDY/YMD	Letter
UK, Ireland, AU, NZ	,	.	,	DMY	A4
Germany, Austria, Belgium, Netherlands, Italy, Spain, Portugal, Denmark, Brazil	;	,	.	DMY	A4
France, Poland, Czech Rep., Norway, Latvia, Estonia	;	,	(space)	DMY	A4
Finland, Sweden	;	,	(space)	DMY/YMD	A4
Switzerland	;	,	'	DMY	A4
Ukraine, Russia	;	,	(space)	DMY	A4
Turkey	;	,	.	DMY	A4
Japan, South Korea	,	.	,	YMD	A4
Mexico	,	.	,	DMY	Letter
Lithuania	;	,	(space)	YMD	A4

Override Dropdowns

Below the country selector, four override dropdowns let you change individual settings without switching countries:

- **Paper** — Letter or A4 (affects PDF report page size)
- **CSV sep** — comma or semicolon (used when importing CSV files)
- **Decimal** — period or comma (used in number display on plots and axes)
- **Date** — MDY, DMY, or YMD (controls the order in which date formats are tried when parsing date columns)

When you change the country, all overrides reset to that country's defaults. Changing an override only affects that one setting.

Saving Defaults

Click **Save as my default** to store your locale preferences globally. These defaults apply to all future sessions regardless of which data file you load. Project settings (target, predictors, parameters) are saved separately in the project database, scoped per project and purpose, via the **Save current as default** buttons; locale defaults persist across all projects via the same database. This two-level approach is designed for organizations like audit firms that work with data from multiple countries — set your most common country as the default, then override per-file when needed.

> ***Tip:*** *Number formatting on plot axes and slope labels automatically adapts to your locale. German locale uses periods for thousands (200.000), Finnish uses spaces (200 000), Swiss uses apostrophes (200'000). No currency symbols are displayed — earthUI is currency-agnostic.*

CHAPTER 4
Appraisal Mode

When you select **For Appraisal** as the Purpose, earthUI configures itself for single-property valuation. All features described in Chapter 3 remain available; this chapter covers only the appraisal-specific additions.

Subject Row Handling

In appraisal mode, **row 1 of your dataset is the subject property** and all remaining rows are comparable sales. Your input file must be organized accordingly (see Chapter 2). The subject's sale price can be left blank or set to any value — earthUI automatically treats it as NA during fitting.

After importing, the Data tab splits into two sections: **Subject Property** (row 1) and **Comparable Sales** (rows 2+). Row 1 is always excluded from model fitting — the notification "Skipping row 1 (subject). Fitting on N rows." confirms this. After fitting, the model still generates predictions for the subject row, shown as `est_<target>` in the output.

In both tables each cell is kept to a single line and truncated to the column width, so wide free-text fields (such as property remarks) do not stretch a row down the page. **Double-click any cell** to open a pop-up showing its full, untruncated contents.

Effective Date & Sale Age

An **Effective Date** field appears in the Variable Configuration section in every purpose mode (defaulting to today's date), since general/CMA and market valuations have an effective date too. It is only *used* to compute `sale_age` in appraisal and market modes. If you designate a column as `contract_date` in the Special column dropdown, earthUI computes a `sale_age` column — the number of integer days between each sale's contract date and the effective date. This column replaces the original date column as a predictor.

The first time you click Fit after designating a contract date, earthUI creates the `sale_age` column and notifies you to click Fit again to include it.

Special Column Designations

In appraisal and market modes, a **Special** dropdown appears for each predictor in the Variable Configuration table. The complete list of special types and their effects:

Special Type	Effect
`no`	Default — no special handling
`contract_date`	Triggers automatic `sale_age` computation from the Effective Date
`listing_date`	Used with contract date to compute DOM when no `dom` column exists
`dom`	Identifies the Days on Market column; displayed in Sales Grid
`concessions`	Identifies sale concessions; used in Net SP formula (Sale Price — Concessions) in Sales Grid
`latitude`	Values rounded to 3 dp; used for Haversine proximity and Location group in Sales Grid
`longitude`	Values rounded to 3 dp; used for Haversine proximity and Location group in Sales Grid
`area`	Grouped in "Loc: Long | Lat | Area" row in Sales Grid
`living_area`	Enables per-SF residuals (`residual_sf`, `cqa_sf`) in download output
`lot_size`	Grouped in "Site Size | Dimensions" row in Sales Grid
`site_dimensions`	Grouped with lot size in Sales Grid
`actual_age`	Grouped in "Actual Age | Effective Age" row in Sales Grid
`effective_age`	Grouped with actual age in Sales Grid
`display_only`	Column included in exports but excluded from model fitting (multiple allowed)

Only one column per special type is allowed (except `display_only`, `remarks`, and `list`, which allow multiple columns — the latter two are advanced roles for the optional Prolog step, see Chapter 6). Assigning a single-column special to a second column automatically clears it from the first. A small blue badge appears next to the variable name showing its assigned special type.

RCA Adjustments Overview

The **Calculate RCA Adjustments & Download** button (sidebar section 7, visible only in appraisal mode after fitting) computes market-derived adjustments for each comparable relative to the subject. The full RCA workflow is described in Chapter 10.

After computing RCA adjustments, the **Generate Sales Grid & Download** button (sidebar section 8) becomes available. The Sales Grid workflow is described in Chapter 11.

> ***Tip:*** *The CQA score you assign to the subject controls how much of the residual distribution is attributed to the subject. A score of 5.00 places the subject at the median of the comparables.*

CHAPTER 5

Market Area Analysis Mode

When you select **Market Area Analysis** as the Purpose, earthUI provides the same real estate–specific features as appraisal mode (special columns, sale age, coordinate rounding) but is oriented toward analyzing a group of properties rather than valuing a single subject.

Differences from Appraisal Mode

- **Skip first row is optional** — a checkbox labeled "Skip first row (subject property)" appears below the Purpose selector. When checked, row 1 is excluded from fitting (like appraisal mode). When unchecked, all rows are included.
- **No RCA or Sales Grid sections** — the "Calculate RCA Adjustments & Download" and "Generate Sales Grid & Download" steps are not available. Market mode focuses on model fitting and output, not per-comparable adjustments.
- **Sidebar numbering** — without the RCA and Sales Grid steps, the Generate Quarto Report section is numbered 7 instead of 9.

When to Use Market Mode

Market Area Analysis mode is appropriate when you are:

- Building a regression model for a neighborhood or market area to understand value drivers
- Analyzing how variables like square footage, age, lot size, and location affect sale prices across a group of properties
- Preparing support for a market conditions analysis or neighborhood delineation
- Working with a dataset that includes a subject property in row 1 but you want the option of including or excluding it from the fit

> ***Tip:*** *Market mode is also useful for general real estate regression where you want special column features (sale age, coordinate rounding) but do not need the RCA adjustment workflow.*

CHAPTER 6

Variable Selection

Section 3 of the sidebar — **Variable Configuration** — is where you choose which columns participate in the model and how they are treated.

Target Variable(s)

The **Target (response) variable(s)** dropdown at the top of Section 3 lists every column in your dataset. Select one column for a standard single-response model, or multiple columns for a multi-response model. When multiple targets are selected, the model fits all responses simultaneously using `earth(cbind(y1, y2, ...) ~ .)`.

Columns selected as targets are automatically excluded from the predictor list.

The Predictor Table

Below the target selector, a table lists every remaining column with the following fields:

Column	Description
Variable	Column name (full name shown in tooltip if truncated). In appraisal/market modes, a blue badge shows the assigned special type.
Type	Data type dropdown: `numeric`, `integer`, `character`, `logical`, `factor`, `Date`, `POSIXct`
Include	Checkbox — include this column as a predictor in the model
Factor	Checkbox — treat this column as a categorical variable
Linear	Checkbox — force linear entry only (no hinge functions)
Latent	Checkbox — hold the column out of the model, reserving it for latent-variable work. Latent-flagged columns are excluded from earth and from the data exported to glmnetUI / mgcvUI
Special	Dropdown (appraisal/market only) — see Special Column Types Reference below
T/F/NA	Three counts per column: logicals show TRUE/FALSE/NA, numerics show $≠0/=$0/NA, other types show 0/0/NA. Shown in red when more than 30% of values are missing

Each row also has a × **/** + **toggle** beside the column name: the red × *soft-disables* a column — it greys out and is held out of the model, but is not deleted — and flips to a green + to re-enable it. Columns are never destroyed.

A hint line above the table explains the abbreviations: "Type = column data type, Inc = include as predictor, Factor = treat as categorical, Linear = linear-only (no hinges)."

Data Type Detection & Overrides

earthUI automatically detects data types on import. Numeric, integer, logical, factor, and date columns are recognized. Character columns that look like dates (at least 80% of values parse against common date formats) are classified as `Date`.

You can override any detection by changing the **Type** dropdown. When you change a column to `character` or `factor`, the Factor checkbox is automatically checked. Changing types affects how the column is passed to the `earth()` function.

Factor and Linear Flags

- **Factor** — When checked, the column is treated as a categorical variable. This is appropriate for columns like `style`, `area_id`, or `grade` that represent discrete groups rather than continuous measurements. Factor columns enter the model as indicator (dummy) variables.
- **Linear** — When checked, the column is forced to enter the model linearly — no hinge (piecewise-linear) functions are created for it. This is useful when you know a variable has a strictly linear relationship with the target.

Special Column Types Reference

In appraisal and market modes, the **Special** dropdown provides the following options. Each type can be assigned to at most one column (except `display_only`, `remarks`, and `list`, which allow multiple):

Date & Time Types:

- `contract_date` — Triggers automatic `sale_age` computation. The original date column is replaced by an integer column measuring days between the sale date and the Effective Date.

- `listing_date` — Used as a fallback for computing Days on Market (DOM = contract date — listing date) when no explicit `dom` column is designated.
- `dom` — Identifies the Days on Market column. Displayed in the Sales Grid's APN row and Date of Sale row.

Monetary Types:

- `concessions` — Identifies sale concessions (seller credits, buyer incentives, etc.). Used in the Sales Grid to compute Net Sale Price: Net SP = Sale Price — Concessions.

Size & Location Types:

- `latitude` — Values are automatically rounded to 3 decimal places to prevent overfitting. Used for Haversine proximity calculations (distance from subject to each comp) and grouped in the Location row of the Sales Grid.
- `longitude` — Same rounding treatment as latitude. Used for proximity and the Location group.
- `area` — Typically a neighborhood or area identifier. Grouped with latitude and longitude in the "Loc: Long | Lat | Area" row of the Sales Grid.
- `living_area` — Enables per-square-foot residual calculations (`residual_sf` and `cqa_sf`) in the download output.
- `lot_size` — Grouped in the "Site Size | Dimensions" row of the Sales Grid.
- `site_dimensions` — Grouped with lot size in the Sales Grid (e.g., "75x120").

Age Types:

- `actual_age` — Grouped in the "Actual Age | Effective Age" row of the Sales Grid.
- `effective_age` — Grouped with actual age in the Sales Grid.

Display Types:

- `display_only` — The column is included in Excel exports but excluded from model fitting entirely. Use this for address fields, MLS numbers, parcel IDs, or other reference data that should not be a predictor. Multiple columns can have this designation.

Text Extraction Types (advanced, optional):

- `remarks` — A free-text column (e.g. `public_remarks`) to be parsed by the optional Prolog step into `pr_*` / `ar_*` feature columns.
- `list` — A comma-delimited column (e.g. `existing_structures`) to be split into one-hot <col>_<value> TRUE/FALSE columns.

The `remarks` and `list` roles do nothing until the optional Prolog / list column processing step is enabled in Settings (see "Advanced: Prolog Remarks Processing" below); multiple columns may carry each of them.

Advanced: Prolog Remarks Processing (Optional)

This feature is **off by default** and entirely optional — earthUI's normal workflow does not depend on it, and it assumes working knowledge of Prolog. Enable it via **Settings (gear icon) → Enable Prolog / list column processing**; the choice is remembered per project.

When enabled, an unnumbered **Execute Prolog Processing** section appears in the sidebar between sections 4 and 5, with a two-pass workflow:

1. **Expand lists & remarks** — parses `remarks` columns with a Prolog DCG grammar into feature columns such as `pr_condition_score`, `pr_has_workshop`, or `pr_garage_area`, and splits `list` columns into one-hot columns. The pass is incremental: only columns designated since the last expansion are processed. Remarks parsing requires the optional **vProlog** package (a self-contained Trealla Prolog engine, not on CRAN); without it, remarks parsing is skipped and `list` splitting (pure R) still runs.
2. **Execute derivation rules** — applies your own Prolog `derive(Col, Val)` rules over the expanded columns (helpers `count_true` / `count_false` / `cell_or` are provided; `drop(Col)` removes a column). Rules live in a per-project <project>_rules.pl file, edited in a floating, dockable code editor, and are syntax-checked and dry-run validated before running.

After either pass, the augmented data frame is written to the project's input folder as <original_name>_<YYMMDD_HHMMSS>.xlsx and becomes the project's **active** data file — reopening the project loads the augmented data without recomputing, and the same file is consumed by glmnetUI and mgcvUI. Generated columns you do not want in the regression can be flagged **Latent** or soft-disabled with the × toggle.

Market area profiles (climate-region priors). Three companion R helpers — `climate_region_for()`, `market_area_profile()`, and `climate_feature_priors()` — classify a project's California location (county + city) into one of seven buyer-recognized climate regions and return ordinal contributory-value priors for HVAC/cooling features (heating, A/C, swamp cooler, whole-house fan, filtration) by region. A companion Prolog module (`inst/prolog/climate_regions_ca.pl`) mirrors the same classification for use inside derivation rules. These priors are experimental and intended to inform valuation of thin-data features and sanity-check estimated coefficients; region assignments should be reviewed by the appraiser.

Multiple Targets

Selecting more than one target variable fits a multi-response Earth model. When multiple targets are selected:

- The model predicts all responses simultaneously, sharing basis functions across targets
- The **wp (response weights)** button becomes active, allowing you to assign a numeric weight to each target
- Variance models (`varmod.method`) are disabled (not supported for multi-response)
- Results tabs display per-response metrics, equations, and diagnostic plots

> ***Tip:*** *Columns designated as* `display_only` *remain in your dataset and appear in the Excel export, but are excluded from the predictor table entirely. Use this for ID columns, addresses, or other reference fields.*

CHAPTER 7

Parameter Selection

Section 4 of the sidebar — **Earth Call Parameters** — provides access to all arguments accepted by the `earth()` function. Each parameter has a blue help icon (**?**) with a tooltip explanation. Parameters are organized into collapsible subsections.

Basic Parameters

Parameter	Default	Description
subset	(empty)	Row filter expression. See "Subset Filtering" below.
weights	NULL	Column selector for case (row) weights. Only numeric columns are listed.
wp	NULL	Response weights for multi-target models. Button opens a dialog with one numeric input per target (default 1.0 each).
keepxy	off	Retain x, y, subset, and weights in the model object.
trace	0	Trace level for fitting output (0 through 5).
glm	none	Optional GLM family: `gaussian`, `binomial`, or `poisson`.
degree	1	Maximum interaction order. Setting degree ≥ 2 auto-enables cross-validation and reveals the Allowed Interactions matrix.
penalty	2	GCV penalty per knot. Higher values produce simpler models.

Forward Pass

Parameter	Default	Description
nk	auto	Maximum terms before pruning. Default: min(200, max(20, $2 \times$predictors)) + 1.
thresh	0.001	Forward-step threshold. Smaller values allow more terms.
minspan	0 (auto)	Minimum span between knots. Negative values set the maximum knots per predictor.
endspan	0 (auto)	End span — minimum distance from a knot to the edge of the data.
newvar.penalty	0	Penalty for introducing a new variable (encourages reuse of existing predictors).
fast.k	20	Number of parent terms to consider in the fast MARS algorithm.
fast.beta	1	Controls the fast MARS aging factor.

Allowed Interactions

When **degree** is set to 2 or higher, an interaction matrix appears below the basic parameters. This is an upper-triangular grid of checkboxes, one for each predictor pair. A checked box means the two predictors are allowed to interact; unchecking it forbids that specific interaction.

Clicking a predictor name (row or column label) toggles all interactions for that variable. **Allow All** and **Clear All** checkboxes at the top provide bulk control. The matrix uses sticky headers so column and row labels remain visible when scrolling.

An info alert reminds you: "Interaction terms increase the risk of overfitting. Cross-validation has been enabled (10-fold)."

Pruning

Parameter	Default	Description
pmethod	backward	Pruning method: `backward`, `none`, `exhaustive`, `forward`, `seqrep`, or `cv`.
nprune	NULL	Maximum terms after pruning. Leave empty for automatic selection.

Cross-Validation

Parameter	Default	Description
nfold	10	Number of cross-validation folds. Set to 0 to disable CV.
ncross	20	Number of cross-validation repetitions.
stratify	on	Stratify CV samples so each fold has a similar response distribution.

When degree $\geq$ 2, cross-validation is automatically enabled (nfold set to 10) to help guard against overfitting from interactions.

Subset Filtering

The **subset** text input accepts an R expression that filters which rows are used for model fitting. You can type an expression directly (e.g., `sale_age < 365 & area_id == 460`) or use the **Build filter…** button to construct one visually.

Build Filter Dialog — Click "Build filter…" to open a guided dialog. Each condition row has a column dropdown, operator (`<`, `>`, `<=`, `>=`, `==`, `!=`), and a value input that adapts to the column type: numeric input for numbers, date picker for dates, and dropdown of unique values for character/factor columns. Conditions are joined with AND (`&`) or OR (`|`) connectors. A preview at the bottom shows the expression and how many rows match. Click **Apply** to insert the expression into the text input.

Date columns must use `as.Date("...")` or `as.POSIXct("...")` wrappers in manual expressions. The Build Filter dialog handles this automatically.

Subset filtering is non-destructive — excluded rows remain in the dataset and receive predictions in the Excel export.

> ***Tip:*** *Use parentheses to group OR conditions when combining with AND. Without parentheses,* `A & B | C` *is evaluated as* `(A & B) | C`*, which may not be what you intend.*

Recommended Parameter Values

earthUI displays a recommended value below key parameters in Section 4. These recommendations update reactively based on the number of fitting rows (n) and selected predictors (p). The formulas are derived from Friedman's MARS paper, earth's internal algorithms, and empirical testing.

Forward Pass Parameters

nk (max terms before pruning):

$$\text{nk} = \text{min}\big(100,\ \text{max}(21,\ 2p+1,\ \lfloor n/10 \rfloor)\big)$$

n	p=5	p=10
30	21	21
50	21	21
100	21	21
200	21	21
500	50	50
1000	100	100
1500	100	100

Earth's default, $\min(200, \max(20, 2p)) + 1$, does not account for dataset size and can be too small for large datasets, constraining the forward pass before it explores all predictors adequately. The $\lfloor n/10 \rfloor$ term allows approximately one term per 10 observations — a standard rule of thumb for avoiding overparameterization. The cap of 100 prevents diminishing returns.

minspan (minimum observations between knots):

$$\text{minspan} = \min\!\left(16,\ \lfloor 5 + n/50 \rfloor\right)$$

endspan (minimum observations from data boundaries):

$$\text{endspan} = \min\!\left(16,\ \lfloor 5 + n/28 \rfloor\right)$$

n	minspan	endspan
30	5	6
50	6	6
100	7	8
200	9	12
300	11	15
500	15	16
1500	16	16

Earth's auto-calculated minspan uses Friedman's equation 43: $\lfloor(-\ln(-\ln 0.95) + \ln(p \cdot n))/(2.5 \ln 2)\rfloor$, which scales as $\ln(np)$ and grows too slowly for large datasets. At $n{=}1500$ it yields only ≈ 7, giving roughly 245 candidate knot locations per continuous predictor — too many, allowing the forward pass to fit noise. The recommended formula targets approximately 100 candidates per predictor for large n, while deferring to earth's auto-calculation for small n (where it is well-tested). Endspan is set slightly larger than minspan to provide additional boundary protection, which helps when data has thin tails (common in real estate).

penalty (GCV penalty per knot):

$$\text{penalty} = \begin{cases} 3 & \text{if degree} > 1 \\ 2 & \text{if degree} = 1 \end{cases}$$

This is earth's own default — 2 for additive models, 3 for interaction models (the higher penalty compensates for the larger search space).

newvar.penalty (penalty for introducing a new predictor):

Recommended: **0.1** when predictors are correlated (e.g., living area with bedroom count, bathroom count, lot size). This biases the forward pass toward reusing predictors already in the model rather than introducing correlated alternatives. The mechanism: each new predictor's RSS improvement is multiplied by $1/(1 + \text{penalty})$. With newvar.penalty $= 0.1$, a new variable must improve RSS by at least 10% more than another knot on an existing variable. This produces simpler models with fewer predictors without affecting final coefficient estimates (the penalty is removed after selection). Leave at 0 if predictors are not correlated.

Pruning Parameters

pmethod: Recommended: **backward**. The default GCV-based backward pruning is deterministic — the same data always produces the same model, with no dependence on random seeds. This is important for reproducibility, especially in appraisal work. The `cv` method uses cross-validation for pruning which introduces seed dependence.

nprune: Recommended: **leave empty** (NULL). Let GCV select the optimal number of terms. Setting nprune imposes a hard cap that overrides GCV's judgment.

Cross-Validation Parameters

nfold (CV folds) — recommended from a banded sample-size table:

n (fitting rows)	nfold
0–299	5
300–399	6
400–499	7
500–599	8

n (fitting rows)	nfold
600–699	9
700+	10

More folds train each fold on more data (a less biased estimate) but shrink the held-out test fold. The table graduates up to the standard 10-fold as the sample grows while keeping each test fold large enough ($≈$50+ rows) for a stable per-fold metric, and falls back to 5-fold for small samples. It caps at 10 — beyond that adds compute (each fold is a full earth fit) for negligible gain. For small samples the recommendation also suggests raising ncross (e.g. 15–20) to stabilise the CVR^2 estimate. With `pmethod = "backward"`, cross-validation does not affect the model — it only computes the diagnostic CVR^2 and provides residuals for the variance model.

ncross (CV repetitions) — also recommended from a banded table:

n (fitting rows)	ncross
0–199	20
200–399	16
400–699	12
700–1199	10
1200–2999	7
3000+	5

Repeats average out fold-partition luck to stabilise CVR^2; larger samples need fewer repeats because a single CV pass is already more stable.

Variance Model

varmod.method: Recommended: **lm** — Milborrow's default: a robust linear variance model (spread is estimated from noisy residuals, so a simple model generalises better). If lm fails to converge on your data, try rlm or const (homoscedastic, always converges), or fall back to earth (also handles a nonlinear spread). A variance model requires `nfold > 1` and `ncross > 1`.

Earth Output File

After every successful fit, earthUI writes `<filename>_earth_output_<timestamp>.txt` to the output folder. This file contains the model terms, summary statistics (R^2, GRSq, CVRSq), variance model details, and trace log. One file is created per fit, providing a cumulative record for comparing parameter configurations.

Saving Parameters

Click **Save current as default** at the bottom of Section 4 to save the current earth parameters **and the Allowed Interactions matrix** to the project (scoped per project and purpose). This is a deliberate, button-only action — nothing is saved as you adjust parameters, and fitting does not save them either, so you can experiment freely. (The companion button in Section 3 saves the target, effective date, and predictor configuration.)

A radio button at the top of Section 4 controls how the parameters initialize: restore the project's last saved parameters, apply your saved defaults, or **Earth defaults** (reset every parameter to `earth()`'s factory values).

CHAPTER 8

Fitting the Earth Model

The Fit Button

Section 5 of the sidebar contains a **Random seed** field followed by the blue **Fit Earth Model** button. The seed controls the cross-validation fold assignments, making fits reproducible: it is pre-filled with a random value on launch, a dropdown below the field recalls recently used seeds, and the seed used is recorded in the Earth Output after fitting. Before fitting, earthUI validates your configuration — a target must be selected and at least one predictor must be included. In appraisal/market modes, latitude and longitude columns are rounded, and sale age is computed from the effective date and contract date (if designated).

Fitting Modal & Trace Output

When you click Fit, a dark modal overlay appears with:

- **Header** — "Fitting Earth Model" with an elapsed timer counting seconds
- **Abort button** — cancels the fit; the background process is stopped and "Aborted by user." is logged
- **Trace log** — a scrollable, monospace text area showing real-time output from the `earth()` function: dataset info, forward pass progress, cross-validation folds, and completion time
- **Color coding** — default lines in green, cross-validation lines in yellow, errors in red

earthUI fits models asynchronously using `callr::r_bg()`, which runs the computation in a separate R process. This keeps the application responsive during long-running fits. A 300ms polling observer reads output from the background process and streams it to the trace log.

When fitting completes, a "Done in X.Xs" message appears and a close button (X) is added to the modal, so you can dismiss it immediately – result tabs render on demand as you click them. The modal then **auto-dismisses** on its own once all outputs have finished rendering.

On success, a green checkmark is appended to the Fit button. If an error occurs, the error message is displayed in the trace log and a notification appears.

Fit Log

Every fit (success or failure) writes a log file to your output folder:

- **Filename**: `<datafile>_earth_log_<YYYYMMDD_HHMMSS>.txt`
- **Contents**: timestamp, all trace output from the fitting process, and any error messages
- **Location**: the active project's output folder (derived automatically from the project)

> ***Tip:*** *If callr is not installed, earthUI falls back to synchronous fitting with a simple progress bar. Install callr for the full trace-output experience.*

CHAPTER 9
Downloading Data

After fitting, download an Excel file with predictions and diagnostics. This output is used in Step 7 (RCA) to assign a CQA (Condition-Quality-Appeal) rating to the subject property. The output is sorted by `residual_sf` and `cqa_sf` to help you assess where the subject falls in the ranking. If the model is good quality, then the properties should be ranked from lowest appealing to most appealing based on residual features that did not go into the regression. The middle value should be approximately 0, the lower half negative values and the upper half positive values. You should find the worst quality homes, or "fixers" near the bottom of the ranking and the nicest homes at the top. There will usually be exceptions for anomalies such as foreclosures, short sales, probate (inheritance related) sales, and quick sales needed for job change or other reasons. Investigation of anomalies usually turns up a pertinent reason for the price anomaly.

The button label adapts to the purpose mode:

- General/Market: **Download Output (Excel)**
- Appraisal: **Download Intermediate Output (Excel)**

The filename format is `<datafile>_modified_<YYYYMMDD_HHMMSS>.xlsx`.

Output Columns

For each target variable, the following columns are appended:

Column	Description
`est_<target>`	Model prediction, e.g. `est_sale_price` (1 decimal place)
`residual`	Actual minus predicted (1 dp)
`cqa`	Condition-Quality-Appeal score (2 dp, range 0–10)
`residual_sf`	Residual divided by living area (if designated, 1 dp)
`cqa_sf`	CQA calculated from ranking via residual_sf (2 dp)
`<variable>_contribution`	Per-g-function contribution value (1 dp, one column per g-function)
`basis`	Intercept value contribution, same for all properties (1 dp)
`calc_residual`	Actual minus (basis + all contributions) — verification column (1 dp)

For multi-target models, a `_<i>` suffix is added to distinguish columns for each target.

Column Ordering & Excel Formatting

The ranking columns are placed at the **leftmost position** in the output file, in this order: `residual_sf`, `cqa_sf`, `residual`, `cqa`. This makes it easy to scan the sorted list and evaluate where the subject falls in the CQA distribution.

These columns have Excel number formatting applied:

Column	Format	Example
`residual_sf`	Numeric, 2 decimal places	12.34
`cqa_sf`	Numeric, 2 decimal places	7.25
`residual`	Numeric, 0 decimal places	15,200
`cqa`	Numeric, 2 decimal places	6.80

CQA Scores

The CQA (Condition-Quality-Appeal) score ranks each row's residual against all other residuals. For a given row, the CQA is the percentage of rows with a smaller signed residual, multiplied by 10. This produces a 0–10 scale where:

- **High CQA** ($\approx$ 9–10) — the property sold for much more than the model predicted (large positive residual)
- **Low CQA** ($\approx$ 0–1) — the property sold for much less than predicted (large negative residual)
- **CQA $\approx$ 5** — the residual is near the median of all residuals

When a `living_area` column is designated, `cqa_sf` provides the same ranking based on per-square-foot residuals.

Sorting & Subject Row

In appraisal and market modes, comparable rows are sorted by `residual_sf` descending (or `residual` if no living area is designated). The subject row (row 1) remains in position 1 and is not sorted. In general mode, rows are exported in their original order.

This sorting, combined with the leftmost ranking columns, allows the appraiser to quickly scan the comparables from most over-predicted to most under-predicted, and assess where the subject's assigned CQA score falls in that distribution.

Factor levels in the prediction data are aligned with the training data before calling `predict()`. Rows with unseen factor levels will produce NA predictions.

> ***Tip:*** *The `calc_residual` column should always equal `residual` (within rounding). If they differ significantly, it indicates a computation issue worth investigating.*

mgcvUI Auto-Export

On every successful fit with `degree` ≤ 2, earthUI automatically saves the full result object as an `.rds` file to the project's output folder. The filename follows the pattern `<datafile>_earthUI_result_<YYYYMMDD_HHMMSS>.rds`.

This file can be loaded by mgcvUI (a companion Shiny app for GAM modeling) using `readRDS()`. mgcvUI uses the earth model's knot locations and basis functions as starting points for GAM smooth terms, enabling a seamless transition from MARS to GAM modeling.

Models with `degree` > 2 are skipped because mgcvUI only supports pairwise interactions. A manual **Export for mgcvUI** button is also available in the sidebar for on-demand export.

glmnetUI Import

The same `.rds` file saved for mgcvUI can also be imported into **glmnetUI** (a companion Shiny app for elastic net regression). In glmnetUI, navigate to **Section 2: Import from earthUI** and use the Browse button to select the `.rds` file.

glmnetUI uses the standard `earth::model.matrix()` approach: the earth model's basis functions (hinges, interactions, and linear terms) become the columns of glmnet's design matrix. glmnet then performs regularized regression on these basis columns, selecting and shrinking them via lasso/elastic net. This combines earth's adaptive basis construction with glmnet's regularization.

Key Benefits

- **Interpretable interactions**: earth's hinge-based interactions (including 3-way terms) have clear geometric meaning, unlike glmnet's native cross-product interactions which are difficult to explain in court or audit settings.
- **Automatic nonlinearity**: earth's hinge functions capture nonlinear relationships that a standard glmnet model with raw predictors would miss.
- **Variable selection**: glmnet can further prune the earth basis, dropping hinge terms that don't contribute.

Important: Weight Column Must Match

> Warning
>
> If earthUI uses a weight column (e.g., to exclude the subject row with weight=0), the same weight column must be designated in glmnetUI's variable configuration. If the weight settings differ, the row counts will not align and the earth basis cannot be appended to glmnet's model matrix.

Workflow

1. Fit an earth model in earthUI (any degree).
2. The `.rds` result file is saved automatically to the project's output folder.
3. In glmnetUI, open **Section 2: Import from earthUI** and browse to the `.rds` file.
4. Verify that the **same data file** and **same weight column** are used in both apps.
5. Configure glmnet parameters in Section 5 (the Interaction Matrix is automatically disabled when an earthUI import is active).
6. Click **Fit Model**. The earth basis columns appear in the Equation, Coefficients, and Contributions tabs.

CHAPTER 10

RCA Calculations & Downloading

The **Calculate RCA Adjustments & Download** button appears in sidebar section 7, visible only in appraisal mode after a model has been fitted. RCA (Residual Constraint Approach) produces market-derived, per-comparable adjustments relative to the subject property.

Opening the RCA Dialog

Clicking the button opens a small modal dialog with:

- **Score type** — radio buttons to choose **CQA** or **CQA per SF** (the SF option is available only when a `living_area` column is designated). If you choose "CQA per SF," then based on the CQA score you assign the subject, its residual score will be its living area times the residual_sf that matches the given CQA_SF score.
- **CQA value** — input for the subject's CQA score (0.00–10.00). The field starts blank, and the Generate button stays disabled until a valid score is entered. This score represents where you believe the subject falls in the quality distribution of the comparables.
- **Generate** button — computes the RCA output and downloads it as Excel

CQA Score Interpolation

When you click Generate, earthUI interpolates the subject's residual from the comparable CQA/residual pairs:

1. The comparables' CQA scores and residuals are sorted by CQA ascending
2. Linear interpolation (`stats::approx()`) maps your entered CQA value to a residual
3. If using CQA per SF, the per-SF residual is converted back to a total residual by multiplying by the subject's living area
4. The subject's estimated value is computed as: model prediction + interpolated residual

Output Columns

The RCA Excel file (`<datafile>_adjusted_<YYYYMMDD_HHMMSS>.xlsx`) includes all intermediate output columns plus:

Column	Description
`subject_value`	Model prediction + interpolated residual (row 1 only)
`subject_cqa`	The CQA score you entered (row 1 only)
`<variable>_adjustment`	Subject contribution minus comp contribution (per g-function)
`residual_adjustment`	Subject residual minus comp residual
`net_adjustments`	Sum of all adjustments (contribution + residual)
`gross_adjustments`	Sum of absolute values of all adjustments
`adjusted_sale_price`	Comp sale price + net adjustments

The adjustment columns tell you, for each comparable, how much the model attributes the difference to each variable. `adjusted_sale_price` is the comparable's sale price after applying all model-derived adjustments — a set of adjusted sale prices that should cluster around the subject's estimated value.

> ***Tip:*** *A CQA score of 5.00 places the subject at the median of the comparables. Scores above 5 indicate the subject is above average quality; below 5 indicates below average.*

CHAPTER 11

Sales Comparison Grid

Overview & Purpose

The **Sales Comparison Grid** is an Excel workbook generated in appraisal mode (sidebar section 8) after computing RCA adjustments. It presents the subject property alongside selected comparables in a structured grid format, with Excel formulas that automatically compute adjustments and an adjusted sale price for each comparable.

The grid is designed for the appraiser's workfile — it combines the regression-derived adjustments from the Earth model with editable cells where the appraiser can allocate the CQA residual to specific property features. The output filename is `SalesGrid_<YYYYMMDD_HHMMSS>.xlsx`.

Comp Selection Modal

Clicking "Generate Sales Grid & Download" opens a modal dialog where you select which comparables to include:

- **Recommended comps** are pre-checked. These are comparables with a gross adjustment percentage below 25% of sale price, sorted by sale age ascending (most recent sales first).
- **Additional comps** are listed below, unchecked by default. These have larger gross adjustments but are still available for selection.
- A maximum of **30 comps** can be selected, producing up to **10 sheets** (3 comps per sheet).

After confirming your selection, the selected comps are sorted by gross adjustment percentage ascending, so the most similar comparables appear on Sheet 1.

Grid Layout

Each sheet has a 20-column layout accommodating the subject property plus 3 comparables. The columns for each entity are:

Entity	Columns (5 per entity)
Subject	Label, Factual Value 1, Factual Value 2, Factual Value 3, Value Contribution (VC)
Each Comp	Factual Value 1, Factual Value 2, Factual Value 3, Value Contribution (VC), Adjustment

The rows from top to bottom are:

1. **Title** — sheet name (e.g., "Intermediate Sales Comparable Grid — Sheet 1 of 3")
2. **Headers** — "Subject" and comp column headers with address
3. **Address** — full property address
4. **APN | MLS# | DOM | Subj.Prox** — parcel number, listing ID, days on market, and Haversine distance (miles) from subject
5. **Sales Price | Concess. | Net SP** — sale price, concessions, and Net Sale Price formula
6. **Regression Features** header row
7. **BASE VALUE** — the model intercept
8. **Date of Sale | OffMkt | OnMkt** — contract date, sale age, and DOM
9. **Grouped rows** (conditional) — Location, Site Size, and/or Age groups
10. **Model variable rows** — one row per predictor (excluding grouped variables)
11. **Blank separator row**
12. **Residual Features** header row
13. **CQA|Residual** — CQA score + remaining residual formula
14. **Residual feature rows** — named + blank rows for appraiser entry
15. **Total VC / Net Adjustment**
16. **Net Adjustment %**
17. **Gross Adjustment %**
18. **Adjusted Sale Price** — formula row
19. **Copyright** footer

Sale Price, Concessions & Net SP

Row 5 shows three values for each comparable:

- **Sale Price** — the comparable's sale price from the data
- **Concessions** — the value from the column designated as `concessions` (0 if not designated)
- **Net SP** — an Excel formula: Sale Price — Concessions

The subject column shows "N/A" for sale price (since it is unknown) and any concessions value if available.

Grouped Rows (Location, Site, Age)

When certain special types are designated and those variables appear in the model, earthUI creates grouped rows that combine related variables:

Loc: Long | Lat | Area — Appears when any of `longitude`, `latitude`, or `area` are in the model. Shows factual values for each constituent variable, a combined Value Contribution (sum of the individual VCs), and for comps, an adjustment (subject combined VC — comp combined VC). Styled with light blue background on VC cells.

Site Size | Dimensions — Appears when `lot_size` or `site_dimensions` are in the model. Same structure as the Location group.

Actual Age | Effective Age — Appears when `actual_age` or `effective_age` are in the model. Same structure.

Variables consumed by grouped rows are **excluded from the individual model variable rows** below, preventing double-counting in the adjustment totals.

Model Variable Rows

Below the grouped rows (or directly below BASE VALUE if no grouped rows), one row per remaining model predictor shows:

- **Factual values** for subject and each comp (the actual data values)
- **Value Contribution (VC)** — the g-function's contribution to the predicted value for that row
- **Adjustment** (comps only) — subject VC minus comp VC

CQA|Residual & Residual Feature Rows

The **CQA|Residual** row shows each property's CQA score and contains a formula for the **remaining residual** — the portion of the residual not yet allocated to specific features. The formula is:

$$\text{Remaining Residual} = \text{Total Residual} - \sum(\text{Residual Feature VCs below})$$

Below this are **residual feature rows** — 5 named rows (Location, View/Appeal, Condition, Quality, Other) plus 6 blank rows. These are input cells where the appraiser enters value contributions for features not captured by the model. As the appraiser fills in values, the Remaining Residual formula automatically decreases.

For each comp, the adjustment column contains a formula: subject feature VC — comp feature VC.

Adjusted Sale Price Formula

The **Adjusted Sale Price** row contains Excel formulas:

- **Subject**: Sum of all Value Contribution cells from BASE VALUE through the last residual feature row. This represents the model's total prediction for the subject plus any appraiser-allocated residual features.
- **Comps**: Net SP + sum of all Adjustment cells from the first grouped or variable row through the last residual feature row. This gives the comparable's sale price after all regression-derived and appraiser-entered adjustments.

Sheet Protection & Editable Cells

Each sheet is **protected** to prevent accidental modification of formulas and data. The protection locks:

- All formula cells (Net SP, remaining residual, adjustments, adjusted sale price, etc.)
- All data cells (addresses, sale prices, factual values, value contributions from the model)
- Labels and headers

The only **unlocked** (editable) cells are the residual feature Value Contribution inputs — the cells under the CQA|Residual row where the appraiser enters breakdowns. These are styled with a light yellow background to indicate they are editable.

Working with the Grid in Excel

1. **Open the file** in Excel or a compatible spreadsheet application.
2. **Review the regression adjustments** — the model-derived adjustments are pre-populated and locked.
3. **Allocate the residual** — in the residual feature rows (yellow cells), enter your assessment of how much of the remaining residual is attributable to each feature (Location, View, Condition, Quality, etc.). The Remaining Residual formula will decrease as you allocate.
4. **Check the Adjusted Sale Price** — this formula automatically updates as you enter residual feature values.
5. **Multiple sheets** — if you selected more than 3 comps, navigate between sheets. Each sheet has the same subject in the left column with 3 different comps.

> ***Tip:*** *The goal is to allocate the CQA residual across the feature rows until the Remaining Residual is close to zero. This demonstrates that you can account for the difference between each comparable's regression estimate and its actual sale price.*

CHAPTER 12

Downloading Reports

After fitting a model, report generation is a **two-step flow** in the sidebar:

1. **Generate Quarto Report** (section 9 in appraisal mode, 7 otherwise) writes a self-contained Quarto bundle — the `.qmd` source plus plots, data, and a reference `.docx` — to the project's output folder. No rendering happens at this step, so the bundle can be edited or combined with other Quarto sources first.
2. **Convert Quarto Report** (section 10) renders any `.qmd` file to the selected output format(s). The source field is auto-filled with the most recently generated bundle, but you can browse to any `.qmd` — useful for converting hand-authored or combined reports.

Report Formats

Three formats are available (one or more can be selected in the Convert section):

- **HTML** — an `.html` file with KaTeX math rendering, embedded images, and a table of contents. Uses the Flatly Bootstrap theme and Roboto Condensed font.
- **Word** — a `.docx` file suitable for editing and distribution. Includes a table of contents, page numbers, and uses a custom reference template for consistent styling.
- **PDF** — typeset with LuaLaTeX for professional-quality output. Paper size follows the locale setting (Letter or A4). Includes landscape pages for large interaction matrices and uses Roboto Condensed with Latin Modern Math for equations.

Rendered files are saved next to the `.qmd` in the project's output folder, and a notification confirms the file locations on completion.

All operations (model fitting, data downloads, RCA calculations, sales grid generation, and report rendering) are logged with start/end timestamps and elapsed times to `<datafile>_earthui_log.txt` in the output folder, for troubleshooting and performance monitoring.

Report Contents

Every report includes the following sections:

1. **Dataset Description** — number of observations, target variable(s), predictors used, categorical predictors
2. **Model Specification** — degree, cross-validation status, number of terms, predictors in model
3. **Allowed Interactions** (degree ≥ 2 only) — the interaction matrix with checkmarks, formatted for the output type
4. **Results Summary** — R^2, GRSq, GCV, RSS, and CV R^2 (if cross-validation was enabled)
5. **Model Equation** — the complete Earth model equation in LaTeX notation
6. **Coefficients & Basis Functions** — table of all terms, their coefficients, and the basis functions that define them
7. **Variable Importance** — bar chart and ranked table of predictor importance scores
8. **g-Function Contributions** — plots for each g-function: line plots for univariate terms, 3D perspective and contour plots for bivariate interactions
9. **Correlation Matrix** — heatmap of predictor correlations
10. **Diagnostics** — Residuals vs Fitted, Normal Q-Q, and Actual vs Predicted plots
11. **ANOVA Decomposition** — table showing each basis function's contribution to RSS
12. **Earth Output** — raw text output from `earth::summary()`, including pruning pass details

For multi-response models, sections 4–8 and 10 are repeated for each target variable.

> ***Tip:*** *Reports require the `quarto` R package. For PDF output, a LaTeX distribution (e.g., TinyTeX) must also be installed.*

CHAPTER 13

Demo Dataset: Appraisal_1.csv

earthUI includes a demo MLS dataset for exploring the appraisal workflow. Load it programmatically with:

```
demo_file <- system.file("extdata", "Appraisal_1.csv", package = "earthUI")
df <- import_data(demo_file)
```

Or import it directly through the Shiny app file upload.

Description

The file contains 1,502 data rows — 1,501 residential sales plus the subject property in row 1 — from a simulated MLS export. The data represents single-family home sales in a multi-area market with a range of property sizes, ages, and locations.

This is not real data, but is based on a realistic neighborhood in Northern California. All identification information has been altered or removed.

Columns

Column	Type	Special Type	Description
weight	numeric	–	Observation weight (0 = exclude from fitting)
id	numeric	display_only	Internal record ID
property_id	numeric	display_only	MLS property identifier
listing_id	character	display_only	MLS listing number
parcel_number	character	display_only	County assessor parcel number (APN)
street_address	character	display_only	Property address
city_name	character	display_only	City
postal_code	character	display_only	ZIP code
county_name	character	display_only	County
contract_date	Date	contract_date	Sale contract date (computes sale_age)
sale_age	numeric	–	Days from contract date to effective date (pre-computed)
coe_date	Date	display_only	Close of escrow date
listing_status	character	display_only	Listing status (e.g., "Sold")
sale_price	numeric	(target)	Sale price — response variable
rent	numeric	–	Monthly rent (for multi-target models)
list_price	numeric	display_only	Listing price
original_list_price	numeric	display_only	Original listing price
living_sqft	numeric	living_area	Gross living area in square feet
beds_total	integer	–	Number of bedrooms
baths_total	numeric	–	Total bath count (e.g., 2.5 = 2 full + 1 half)
lot_size	numeric	lot_size	Lot size in square feet
area_id	integer	area	MLS area identifier
area_text	character	display_only	Area name
age	numeric	actual_age	Property age in years
year_built	integer	display_only	Year of construction
latitude	numeric	latitude	Latitude (rounded to 3 dp for model)
longitude	numeric	longitude	Longitude (rounded to 3 dp for model)
latitude6	numeric	display_only	Full-precision latitude
longitude6	numeric	display_only	Full-precision longitude
garage_spaces	integer	–	Number of garage bays
fp_count	integer	–	Number of fireplaces
no_of_stories	numeric	–	Number of stories
style	character	–	Architectural style
view	character	–	View type (e.g., "Neighborhood", "Hills")
days_on_market	integer	dom	Days on market
listing_date	Date	listing_date	Listing date
sale_concessions	numeric	concessions	Seller concessions

Suggested Quick Start

1. Launch earthUI: `earthUI::launch()`
2. Create (or open) a project with purpose **For Appraisal**
3. Copy `Appraisal_1.csv` into the project's input folder and pick it in Section 2
4. Select `sale_price` as the target
5. Assign special types as shown in the table above
6. Include predictors: `sale_age`, `living_sqft`, `baths_total`, `lot_size`, `area_id` (as factor), `age`, `latitude`, `longitude`, `garage_spaces`
7. Set degree to 1, click **Fit Earth Model**
8. Download intermediate output (Step 6), review the CQA ranking

9. Compute RCA adjustments (Step 7) with a CQA score of $\sim$5.00
10. Generate the Sales Comparison Grid (Step 8)

> ***Tip:*** *The demo dataset is a good way to explore all of earthUI's features before importing your own MLS data. The suggested predictors above typically produce a model with R^2 above 0.85.*